

MSA Competency Matrix - Roles and Required Knowledge Levels

L1=Awareness, L2=Practitioner, L3=Expert | AIAG MSA 4th Edition

QUALITY HEAD AGENTS | AIAG MSA 4th Edition | IATF 16949:2016

Competency Matrix by Role

Quality Head:	L3 - GRR Studies L3 - Bias/Linearity/Stability L3 - Attribute GRR L3 - IATF Audit L3 - PPAP Element 8
Quality Manager:	L3 - GRR Studies L2 - Bias/Linearity/Stability L2 - Attribute GRR L2 - IATF Audit L3 - PPAP Element 8
Quality Engineer:	L2 - GRR Studies L2 - Bias/Linearity/Stability L2 - Attribute GRR L2 - IATF Audit L2 - PPAP Element 8
Metrology Technician:	L3 - GRR Studies L2 - Bias/Linearity/Stability L1 - Attribute GRR L1 - IATF Audit L1 - PPAP Element 8
Operator/Inspector:	L1 - GRR Studies L1 - Stability Monitoring L1 - Attribute GRR L1 - Gauge Use L1 - Reaction

Training Evaluation Methods

Level 1:	Written test (multiple choice, 15 questions). Practical: conduct 5 measurements correctly. Pass mark: 80%.
Level 2:	Written test (30 questions, including calculations). Practical: conduct and interpret a 2-operator GRR study. Pass mark: 75%.
Level 3:	IATF audit simulation: respond to 10 auditor questions. Practical: facilitate full MSA study. Peer review of GRR report. Pass mark: 80%.
Retraining:	Required when: GRR result worsens after staff change, audit NC on MSA, new gauge type introduced, customer complaint linked to measurement error.

Key Competency Definitions

L1 - Awareness:	Understands what the concept is and why it matters. Can follow a procedure. Cannot calculate or design a study.
L2 - Practitioner:	Can plan, conduct, calculate, and interpret the study independently. Can train L1. Understands when action is required.
L3 - Expert:	Can facilitate, audit, advise, and judge edge cases. Can respond to customer/auditor challenges. Can train L2 staff.